



GE Medical Systems

Technical Publication

Direction 2291842-100

Revision 0

**GE Medical Systems
DASM I & II Installation Manual**

Copyrighted© 2001 by GE Medical Systems

<http://www.gemedicalsystems.com>



Published by GE Medical Systems

CTSE Manager

John Close

Program Integrators

Jim Pirkey

Editor-in-Chief

Rick Fauska

Technical Editors and Writers

George Farrington

Tim DallaValle (Compuware)

LEGAL NOTES & PRECAUTIONS

TRADEMARKS

Adobe, the Adobe logo, Acrobat, the Acrobat logo, Exchange, and PostScript are trademarks of Adobe Systems Incorporated or its subsidiaries and may be registered in certain jurisdictions. Microsoft is a registered trademark and Windows is a trademark of Microsoft Cooperation. All other products and their name brands are trademarks of their respective holders.

COPYRIGHTS

All Material Copyrighted (c) 2001 by the GE Company, All rights reserved.

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTAR REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GEMS 以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Revision	Date	Reason for change
0	01/17/01	Initial release of document.

List of Effected Pages

PAGES	REVISION	PAGES	REVISION	PAGES	REVISION
1 through 20					

Table of Contents

Chapter 1

DASM I 9

Section 1.0

Console w/Serial Expander 9

Section 2.0

Console w/o Serial Expander 12

Chapter 2

DASM II 15

Section 1.0

Console w/Serial Expander 15

Section 2.0

Console w/o Serial Expander 17

Chapter 1

DASM I

Two pieces of information are required in order to install a DASM. First, you must know the version of the DASM to be installed: DASM I or DASM II. Second, you must know the configuration of the console hardware: with or without a serial expander.

DASM I uses the installation procedures outlined in chapter 1 (this chapter), while DASM II uses chapter 2. If your console uses a serial expander, then you should use Section 1 of the appropriate chapter. If your console DOES NOT use a serial expander, then you should use Section 2 of the appropriate chapter. All other sections may be ignored.

Section 1.0

Console w/Serial Expander

Note: If the DASM does NOT fit in the console, then place it in the space beneath the console tabletop, or store it behind the console, on the floor.

- 1.) Refer to [Figure 1-1](#).
- 2.) Remove screws (items C) and cover (item A) and set aside.
- 3.) Verify/Remove third jumper pin from ribbon cable (view B).

Note: Follow this step to ensure the DASM is set to SCSI ID 1.

- 4.) Replace cover and secure with screws.

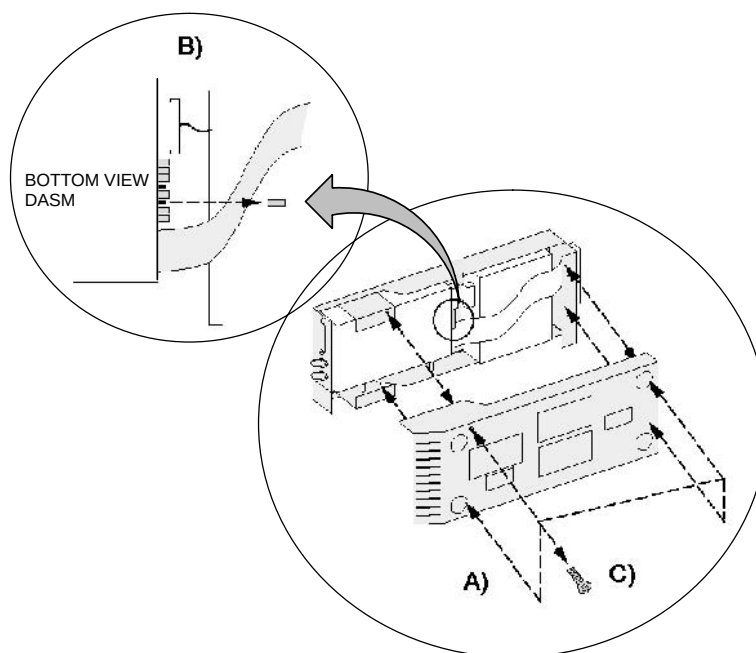


Figure 1-1 DASM - Enlarged Detail

- 5.) Refer to [Figure 1-2](#). Attach cable 46-296980G1 to the front of the DASM.

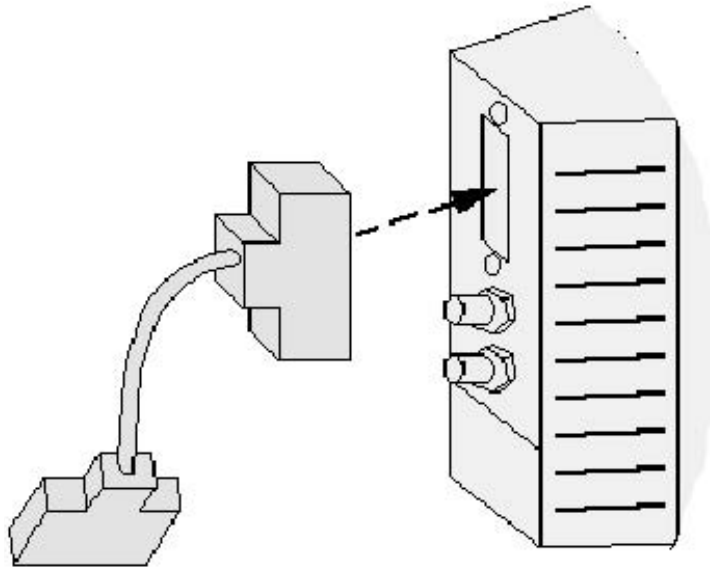


Figure 1-2 Analog DASM Detail

- 6.) At the Central Data Box, disconnect the SCSI cable that comes from the Octane computer. This cable is shown as cable A (2191432-2) in [Figure 1-3](#).

Rear View of Octane and Central Data Box

SCSI Cable Configuration without DASM

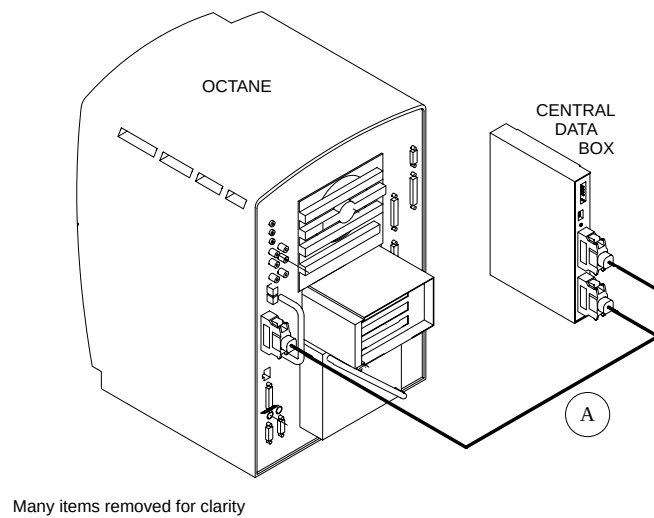


Figure 1-3 SCSI Cable Configuration without DASM

- 7.) Attach the freed end of the SCSI cable (from the Octane) to the DASM, using a SCSI 2 to

SCSI 1 adapter (2209910). Refer to [Figure 1-4](#).

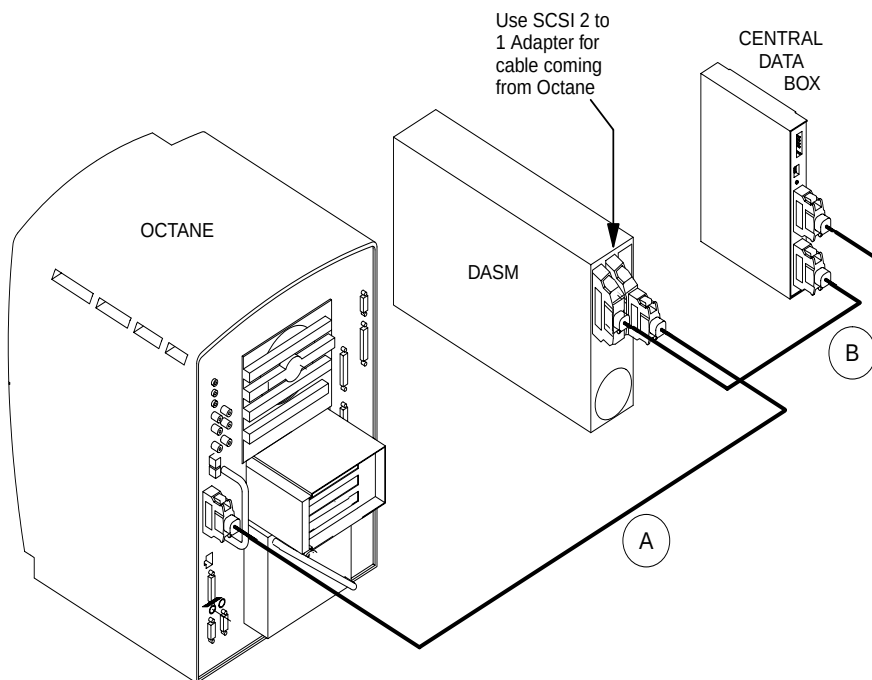


Figure 1-4 SCSI Cable Configuration with DASM

- 8.) Attach SCSI cable B (2110495-2) between the Central Data Box and DASM, as shown in [Figure 1-4](#).
- 9.) Plug DASM power cord into J10 (see [Figure 1-5](#)).

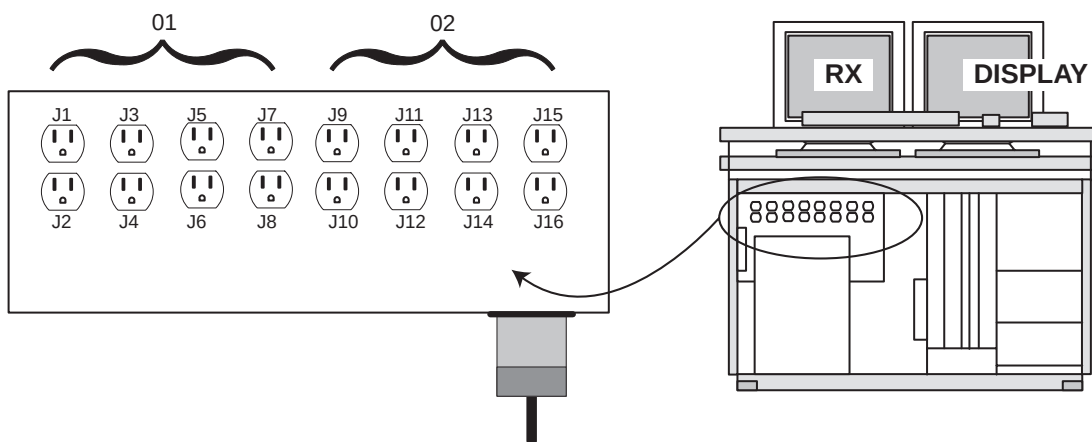


Figure 1-5 Console Power Panel

- 10.) Reattach remaining cable(s) from camera to DASM.

Section 2.0

Console w/o Serial Expander

Note: The DASM does NOT fit in the console. Place the DASM in the space beneath the console tabletop, or store it behind the console, on the floor.

- 1.) Refer to [Figure 1-6](#).
- 2.) Remove screws (items C) and cover (item A) and set aside.
- 3.) Verify/Remove third jumper pin from ribbon cable (view B).

Note: Follow this step to ensure the DASM is set to SCSI ID 1.

- 4.) Replace cover and secure with screws.

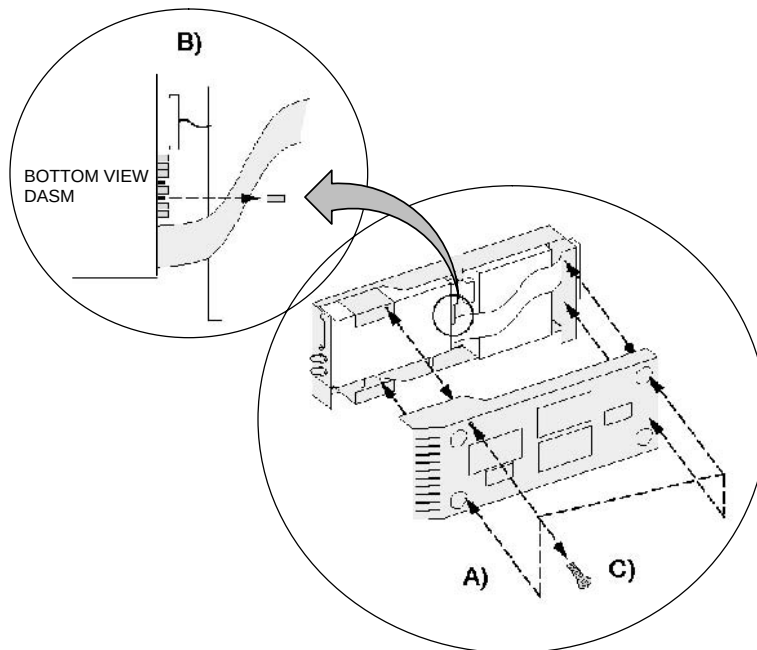


Figure 1-6 DASM - Enlarged Detail

- 5.) Attach a SCSI terminator (46-269601P1) to the bottom SCSI port, as shown in [Figure 1-7](#).



Figure 1-7 DASM SCSI Termination Block

- 6.) Refer to [Figure 1-8](#). Attach cable 46-296980G1 to the front of the DASM.

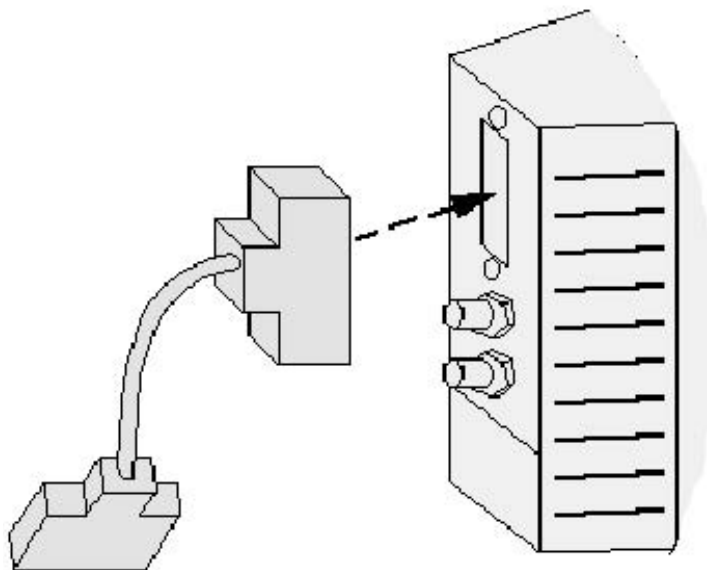


Figure 1-8 Analog DASM Detail

- 7.) Refer to [Figure 1-9](#). Install SCSI 2 to SCIS 1 Adapter (2209910) to the back of the DASM.

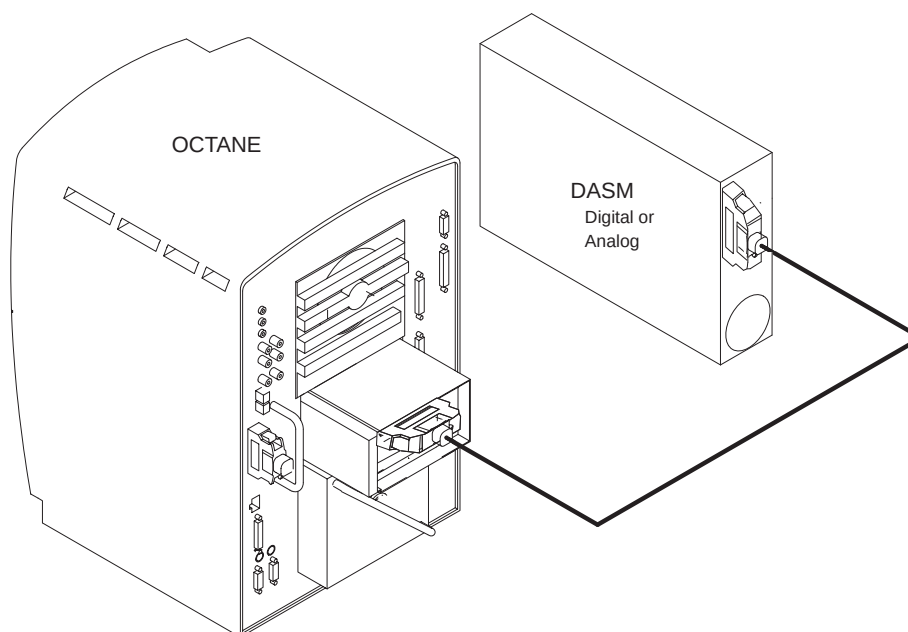


Figure 1-9 SCSI Cable Configuration with DASM

- 8.) Attach the free end of the SCSI cable (2277403) to the PCI module's SCSI connector (slot 1 — top slot).
9.) Plug DASM power cord into J10 (see [Figure 1-5, on page 11](#)).
10.) Reattach remaining cable(s) from camera to DASM.

Chapter 2

DASM II

Two pieces of information are required in order to install a DASM. First, you must know the version of the DASM to be installed: DASM I or DASM II. Second, you must know the configuration of the console hardware: with or without a serial expander.

DASM I uses the installation procedures outlined in chapter 1, while DASM II uses chapter 2 (this chapter). If your console uses a serial expander, then you should use Section 1 of the appropriate chapter. If your console DOES NOT use a serial expander, then you should use Section 2 of the appropriate chapter. All other sections may be ignored.

Section 1.0

Console w/Serial Expander

DASMs are available in two types, digital (refer to [Figure 2-1](#)) and analog (refer to [Figure 2-2](#)). If DASMs are used at your site, they may already be set at the factory. Complete the following steps to connect a DASM to the Octane computer.

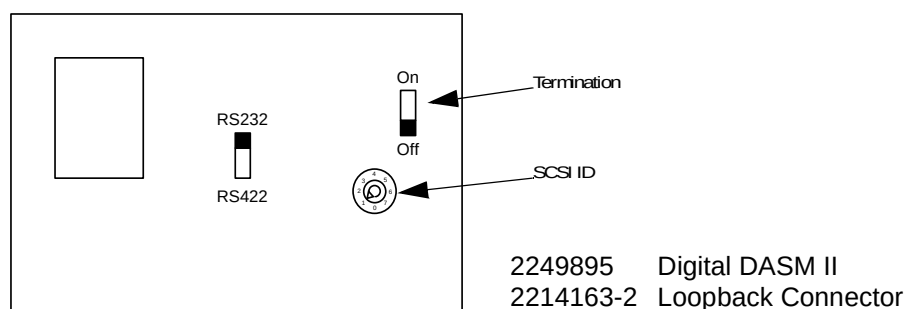


Figure 2-1 Digital DASM Bottom

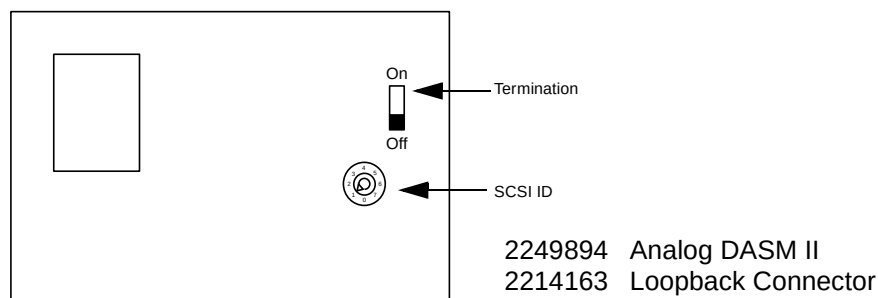


Figure 2-2 Analog DASM Bottom

- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to off.
- 4.) Set the RS232/RS422 switch to the RS232 position if you have a digital DASM.

- 5.) Disconnect the Central Data Box end of SCSI cable A (2191432-2). Refer to [Figure 2-3](#).

Rear View of Octane and Central Data Box

SCSI Cable Configuration without DASM

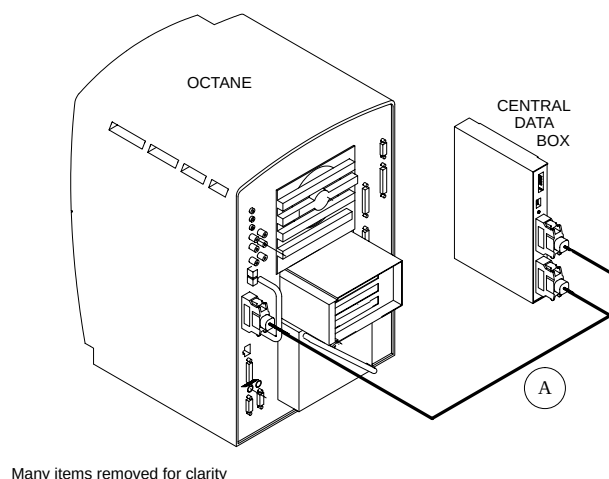


Figure 2-3 SCSI Cable Configuration without DASM

- 6.) Refer to [Figure 2-4](#). Install SCSI 2 to SCSI 1 Adapter (2209910) to the back of the DASM.

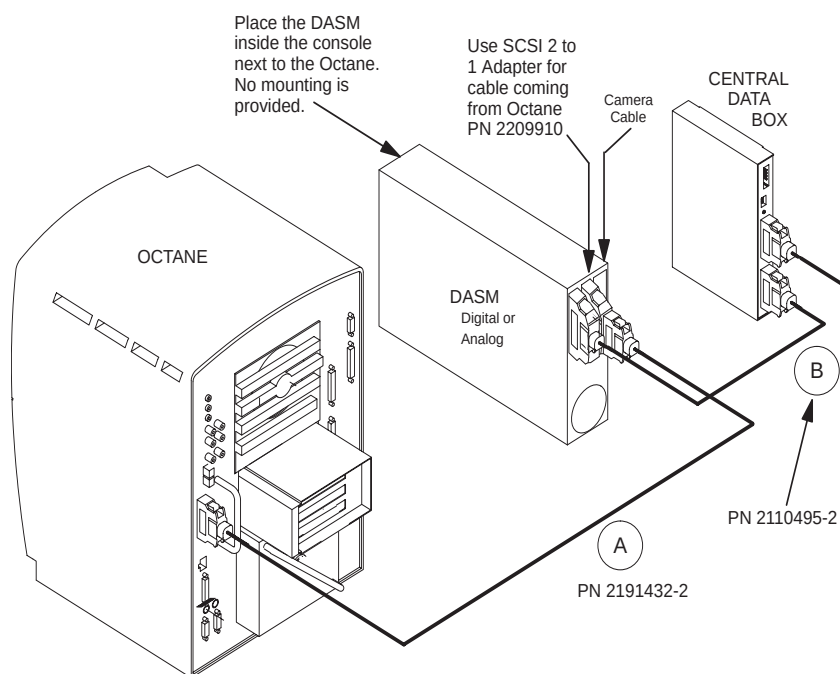


Figure 2-4 SCSI Cable Configuration with DASM

- 7.) Attach the free end of cable A (HD50) to adapter installed in step (6).
8.) Attach cable B between the Central Data Box and DASM as shown.
9.) Plug DASM power cord into J10 (refer to [Figure 1-5, on page 11](#)).

- 10.) Re-attach remaining cable(s) from camera to DASM, and install the DASM next to the computer, as shown in [Figure 2-5](#).

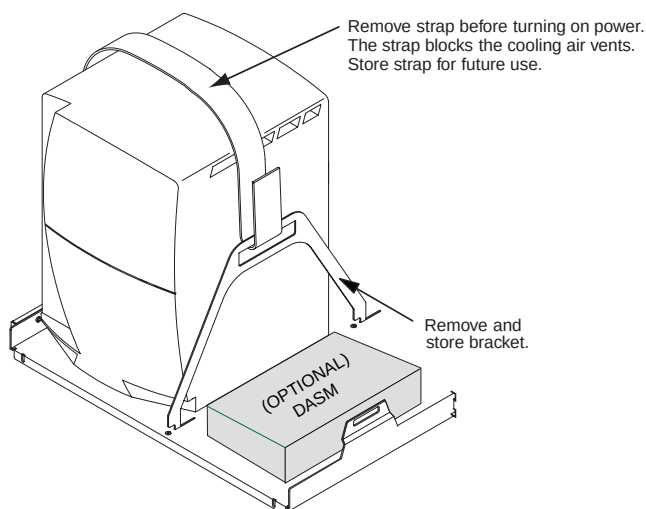


Figure 2-5 DASM Location

Section 2.0

Console w/o Serial Expander

Although DASMs are available in two types (digital and analog), their setup is identical. Complete the following steps to connect a DASM to the Octane computer.

- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to ON.

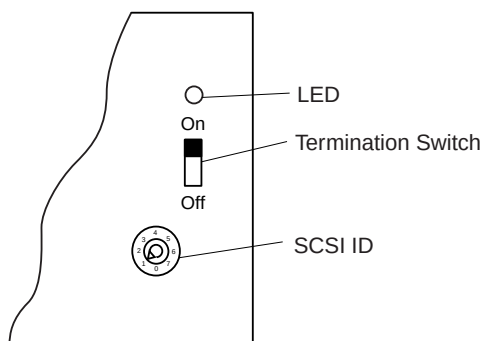


Figure 2-6 DASM Bottom

- 4.) Set the RS232/RS422 switch to the RS232 position, if you have a digital DASM.
- 5.) Refer to [Figure 2-7](#). Attach the SCSI cable (2277403) supplied with DASM kit to the back of the DASM.

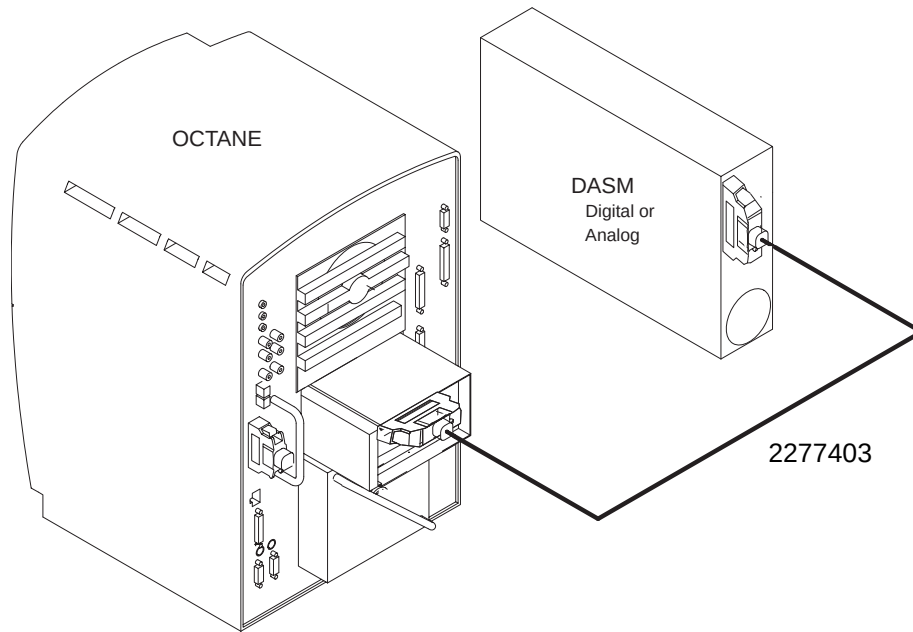


Figure 2-7 SCSI Cable Configuration with DASM

- 6.) Attach the free end of the SCSI cable (HD68) to the PCI module's SCSI connector (slot 1 — top slot).
- 7.) Plug DASM power cord into J10 (refer to [Figure 1-5, on page 11](#)).
- 8.) Attach remaining cable(s) from camera to DASM (refer to [Figure 2-5, on page 17](#)).



GE MEDICAL SYSTEMS

**GE MEDICAL SYSTEMS-AMERICAS: FAX 262.544.3384
P.O. BOX 414; MILWAUKEE, WISCONSIN 53201-0414, U.S.A.**

**GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE**

**GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPOR**

<http://www.gemedicalsystems.com>